

Piter Z. Garcia Bautista

Machine Learning, Diagnostics, and Reproducible Computational Research

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Summary

Graduate researcher with an M.S. in Computer Science and current graduate work in Data Science at Rochester Institute of Technology. Brings industry experience in healthcare AI and analytics, current research experience building reproducible machine learning evaluation systems, and applied coursework in computational biology and bioinformatics. Research interests center on robust and equitable machine learning for diagnostics, decision support, and heterogeneous biomedical data.

Technical Skills

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash

Machine Learning / AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, supervised learning, feature engineering, model evaluation, error analysis, fairness-aware ML

Data Science / Statistics: pandas, NumPy, SciPy, Matplotlib, Plotly, Seaborn, exploratory data analysis, reproducible benchmarking, applied statistics

Research Tooling: Git/GitHub, Linux, Docker, LaTeX, AWS, GCP

Computational Biology: bioinformatics workflows, high-throughput sequencing analysis, differential expression, structured quality control

Relevant Experience

Graduate Assistant – Quantum & AI Research

Fall 2025 – Present

Software Engineering Department, Rochester Institute of Technology

Rochester, NY

- Build reproducible machine learning evaluation workflows across local, Colab, and GCP environments, with shared datasets, structured logging, automated validation, and comparison-ready experiment outputs.
- Develop Python-based experimental systems for model selection and decision-making under noisy, stochastic, and heterogeneous conditions, emphasizing reliability, traceability, and clear technical reporting.
- Produce manuscript-ready analyses and technical artifacts for faculty-led research, with particular attention to reproducibility and disciplined evaluation methodology.

AI Data Solutions Engineer

Jan 2021 – Sep 2022

VIOME Inc.

New York, NY

- Developed applied AI and machine learning workflows in Python for healthcare-oriented use cases, including preprocessing, feature engineering, model experimentation, and evaluation.
- Worked in AWS-backed environments to support reproducible data preparation and reliable evaluation for predictive and recommendation-oriented health applications.

Data Solutions Engineer

Sep 2022 – Aug 2024

VEDADATA Inc.

New York, NY

- Designed large-scale Python and pandas analytics workflows for ingestion, cleaning, preprocessing, validation, and downstream analysis.
- Applied data-quality checks and statistical reasoning to improve traceability, reliability, and usability in production data systems.

Computational Biology / High-Throughput Sequencing Analysis Project

Spring 2026

Rochester Institute of Technology

Rochester, NY

- Completed course-based computational biology work focused on differential expression analysis, quality-control validation, and interpretation of high-dimensional biological data.
- Produced structured reports connecting preprocessing, statistical analysis, and biological interpretation in a reproducible workflow.

Education

Master of Science in Data Science

Expected Aug 2026

Rochester Institute of Technology

Rochester, NY

- Relevant coursework: Data Science Foundations, Software Engineering for Data Science, Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics.

Master of Science in Computer Science

Rochester Institute of Technology

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.

2015

Rochester, NY

Additional Graduate Study: M.S. in Teaching Computer Science K–12

University of Rochester, Warner School of Education

- NSF Robert Noyce Scholar; Advanced Certificate in Leadership in Disability and Inclusive Practices.

Expected Aug 2026

Rochester, NY

Selected Fit for This Role

- Healthcare AI industry experience combined with current graduate work in data science and computational biology.
- Strong Python-based machine learning, evaluation, and reproducibility background aligned with method development on complex and heterogeneous data.
- Long-term research direction focused on trustworthy, equitable diagnostic and decision-support systems.